

Solutions to O'Donnell, Analysis of Boolean Functions, Exercise 2.22

Throughout, for odd n write $n = 2m + 1$ and use the standard majority function $\text{Maj}_n(x) = \text{sgn}(x_1 + \dots + x_n)$ on $\{-1, 1\}^n$. Thus there is no tie when n is odd.

(a) Claim:

$$\text{Inf}_i[\text{Maj}_n] = \text{binom}(n-1, (n-1)/2) \cdot 2^{-(1-n)} \text{ for every } i \text{ in } [n].$$

Proof. Fix i and condition on all coordinates except x_i . Among the remaining $n-1 = 2m$ coordinates, let k be the number of $+1$'s.

If $k > m$, the other coordinates already have a strict positive majority, so changing x_i cannot change Maj_n . If $k < m$, the other coordinates already have a strict negative majority, so changing x_i again cannot change Maj_n . The only possible pivotal case is $k = m$.

When $k = m$, the other $2m$ coordinates are tied. Setting $x_i = 1$ makes the total sum 1, while setting $x_i = -1$ makes the total sum -1 . Thus majority changes. Hence i is pivotal exactly when the remaining $2m$ bits contain exactly m many $+1$'s.

There are $\text{binom}(2m, m)$ such assignments of the remaining coordinates and $2^{(2m)}$ total assignments. Therefore

$$\begin{aligned} \text{Inf}_i[\text{Maj}_{(2m+1)}] &= \text{binom}(2m, m) / 2^{(2m)} \\ &= \text{binom}(n-1, (n-1)/2) \cdot 2^{-(1-n)}. \end{aligned}$$

(b) Claim: $\text{Inf}_1[\text{Maj}_n]$ is decreasing as a function of odd n .

Proof. Let

$$a_m = \text{Inf}_1[\text{Maj}_{(2m+1)}] = \text{binom}(2m, m) / 4^m.$$

Then

$$\begin{aligned} a_{(m+1)} / a_m &= \text{binom}(2m+2, m+1) / (4 \cdot \text{binom}(2m, m)) \\ &= ((2m+2)(2m+1)) / (4(m+1)^2) \\ &= (2m+1)/(2m+2) < 1. \end{aligned}$$

Thus a_m is strictly decreasing in m , equivalently $\text{Inf}_1[\text{Maj}_n]$ is decreasing over odd n .

(c) Claim:

$$\text{Inf}_1[\text{Maj}_n] = \sqrt{2/\pi} / \sqrt{n} + O(n^{-3/2})$$

for odd n .

Proof. From part (a),

$$\text{Inf}_1[\text{Maj}_{(2m+1)}] = \text{binom}(2m, m) / 4^m = (2m)! / ((m!)^2 4^m).$$

Stirling's formula gives

$$m! = \sqrt{2\pi m} (m/e)^m (1 + O(1/m)).$$

Applying this also to $(2m)!$ yields

$$\begin{aligned} \text{binom}(2m, m) &= \sqrt{4\pi m} (2m/e)^{(2m)(1+O(1/m))} \\ &\quad / ((2\pi m)^m (m/e)^{(2m)(1+O(1/m))}) \\ &= 4^m / \sqrt{\pi m} \cdot (1+O(1/m)). \end{aligned}$$

Therefore

$$\text{Inf}_1[\text{Maj}_{(2m+1)}] = 1/\sqrt{\pi m} + O(m^{-3/2}).$$

Since $m = (n-1)/2$,

$$\begin{aligned} 1/\sqrt{\pi m} &= \sqrt{2/\pi} / \sqrt{n-1} \\ &= \sqrt{2/\pi} / \sqrt{n} + O(n^{-3/2}). \end{aligned}$$

The claim follows.

(d) Claim:

$$2/\pi \leq \text{Inf}_1[\text{Maj}_n] \leq 2/\pi + O(n^{-1})$$

for odd n .

Proof. For monotone Boolean functions, Proposition 2.21 gives $\hat{f}(i) = \text{Inf}_i[f]$.

Majority is monotone and symmetric, so all degree-one Fourier coefficients are equal.

Thus

$$W^1[\text{Maj}_{(2m+1)}] = (2m+1) a_m^2,$$

where $a_m = \text{binom}(2m, m) / 4^m$.

Let $B_m = (2m+1)a_m^2$. Using the ratio from part (b),

$$\begin{aligned} B_{(m+1)} / B_m &= ((2m+3)/(2m+1)) * ((2m+1)/(2m+2))^2 \\ &= 1 - 1/(2m+2)^2 < 1. \end{aligned}$$

So B_m decreases with m . Part (c) gives

$$\begin{aligned} B_m &= (2m+1) (\sqrt{2/\pi}) / \sqrt{2m+1} + O((2m+1)^{-3/2})^2 \\ &= 2/\pi + O(n^{-1}). \end{aligned}$$

Thus B_m tends to $2/\pi$. Since B_m is decreasing to this limit, $B_m \geq 2/\pi$ for all m , and the asymptotic estimate gives the upper bound $B_m \leq 2/\pi + O(n^{-1})$.

This is exactly the claimed estimate for $W^1[\text{Maj}_n]$.

(e) Claim:

$$\sqrt{2/\pi} \sqrt{n} \leq I[\text{Maj}_n] \leq \sqrt{2/\pi} \sqrt{n} + O(n^{-1/2})$$

for odd n .

Proof. Total influence is the sum of individual influences, and all coordinates are symmetric for majority:

$$I[\text{Maj}_n] = n \text{Inf}_1[\text{Maj}_n].$$

Using part (c),

$$\begin{aligned} I[\text{Maj}_n] &= n (\sqrt{2/\pi} / \sqrt{n} + O(n^{-3/2})) \\ &= \sqrt{2/\pi} \sqrt{n} + O(n^{-1/2}). \end{aligned}$$

This gives the upper estimate after increasing the implicit constant if needed.

For the lower bound, part (d) gives

$$W^1[\text{Maj}_n] = n \text{Inf}_1[\text{Maj}_n]^2 \geq 2/\pi.$$

Since $\text{Inf}_1[\text{Maj}_n] \geq 0$,

$$\begin{aligned} I[\text{Maj}_n] &= n \text{Inf}_1[\text{Maj}_n] \\ &= \sqrt{n W^1[\text{Maj}_n]} \\ &\geq \sqrt{2/\pi} \sqrt{n}. \end{aligned}$$

Combining the two bounds proves the claim.

(f) Claim: If n is even and $f: \{-1, 1\}^n \rightarrow \{-1, 1\}$ is a majority function, then

$$I[f] = I[\text{Maj}_{(n-1)}] = \sqrt{2/\pi} \sqrt{n} + O(n^{-1/2}).$$

Proof. Write $n = 2m$. A majority function agrees with the sign of $x_1 + \dots + x_{(2m)}$ whenever this sum is nonzero, and may break ties arbitrarily when the sum is zero.

We count boundary edges of the hypercube, i.e. unordered edges whose endpoints have different f -values. Total influence equals

$$I[f] = 2 * \{\text{boundary edges of } f\} / 2^{(2m)},$$

because each boundary edge contributes once from each endpoint to average sensitivity.

A boundary edge must have a unique endpoint on the tie layer

$$x_1 + \dots + x_{(2m)} = 0.$$

There are $\text{binom}(2m, m)$ tie vertices.

Fix a tie vertex x . It has exactly m coordinates equal to $+1$ and m coordinates equal to -1 . If $f(x)=1$, flipping a $+1$ coordinate moves to total sum -2 , where every majority function has value -1 ; these are exactly m boundary edges incident to x . Flipping a -1 coordinate moves to total sum $+2$, where the value is $+1$, so those edges are not boundary edges. If $f(x)=-1$, the same argument with signs reversed again gives exactly m boundary edges incident to x .

Thus every tie vertex contributes exactly m boundary edges, independent of the tie-breaking rule. Hence

$$\{\text{boundary edges}\} = m \text{binom}(2m, m),$$

and

$$I[f] = 2m \text{binom}(2m, m) / 2^{(2m)}.$$

By part (a),

$$I[\text{Maj}_{(2m-1)}] = (2m-1) \text{binom}(2m-2, m-1) / 2^{(2m-2)}.$$

Using

$$\text{binom}(2m, m) = ((2m)(2m-1)/m^2) \text{binom}(2m-2, m-1),$$

we get

$$\begin{aligned} 2m \text{binom}(2m, m) / 2^{2m} \\ &= (2m-1) \text{binom}(2m-2, m-1) / 2^{2m-2} \\ &= I[\text{Maj}_-(2m-1)]. \end{aligned}$$

Therefore $I[f] = I[\text{Maj}_-(n-1)]$. Applying part (e) to the odd integer $n-1$ gives

$$\begin{aligned} I[f] &= \sqrt{2/\pi} \sqrt{n-1} + O(n^{-1/2}) \\ &= \sqrt{2/\pi} \sqrt{n} + O(n^{-1/2}). \end{aligned}$$